

NexGen iMATCH Implementation Plan

The plan followed, phase by phase, with every gate and every outcome — including the ones that failed

This is the implementation record of NexGen iMATCH across a two-month build: the plan as it was written, the order in which it was executed, the gate fixed in advance of each measured run, and the result that gate returned. **Part A** covers the plan as executed, from the pre-repository design work in June 2026 through to 5 August 2026 — including the day the recognition engine was found to be a placeholder and rebuilt, the audit day that overturned three assumptions at once, and the experiment that refuted this project's own central research thesis. **Part B** covers the plan ahead: the phase order, the multi-year stage programme, the data strategy, the risk register, and an explicit list of what will not be built. Failed gates are recorded with the same weight as passed ones, because a programme that only publishes its wins cannot be trusted on its wins.

AUTHOR	Lochan Sharma — sole planner, implementer and author of record
PROJECT	NexGen iMATCH — forensic face recognition and identity intelligence
PERIOD COVERED	June 2026 – 5 August 2026 (version control from 1 July 2026)
DOCUMENT	Part 2 of 2 · Companion volume: System Architecture
STATUS	Current as of 5 August 2026 · plan of record
BASIS	Git history read in full, experiment result artefacts, benchmark and claims records

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SECTION 00

Document control

Author	Lochan Sharma. Sole planner, implementer and author. Every phase in this document was designed, executed and measured by her.
Scope	The implementation plan for NexGen iMATCH: the order of work, the gate attached to each measured decision, and the outcome. Architecture is covered in the companion volume.
Period	June 2026 through 5 August 2026 — approximately two months. The June period precedes version control and covers project design, dataset acquisition and the initial codebase; it enters the record on 1 July 2026 as the first commit's 75 files and 11,051 lines.
Basis	The repository's own git history read in full, the experiment result artefacts, the benchmark record, the claims register, the failure log, and the scorecard.
Reconciliation	Line and commit totals in this document reconcile exactly against git. Where a figure could not be itemised — the pre-repository June period has no commits to break down by day — that is stated rather than estimated.

103 COMMITTS	18 WORK SESSIONS	28 EXPERIMENTS	73 FAILURES LOGGED	351 TESTS	12 DELIVERY DOCS
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SECTION 01

How the plan is run

The plan is not a list of features. It is a sequence of decisions, each attached to a measurement, each with its pass and fail thresholds fixed before the run.

The six rules that govern a result

#	Rule	Why
1	Gates are pre-registered. Pass and fail thresholds are written down before the data is seen.	A gate decided after the result is not a gate. Every experiment in Section 12 and Section 14 carries its threshold in writing from before the run.
2	A negative result is a deliverable.	The project's own central research thesis was refuted, and the refutation is a first-class artefact with its own commit. Three of the largest pieces of work in this record ended in a rejection.
3	Contamination is reported as a floor, never as a total.	An audit finds what it can detect. Reporting a detected overlap as the total understates risk in the direction that flatters the result.
4	Every claim is re-verified after any model, threshold or fusion change.	The claims register mandates it in its own header, and a regression gate enforces it.
5	A measurement that cannot state its protocol is not a measurement.	Sample counts, confidence intervals, reference population and disjointness are stated with every figure or the figure is withdrawn.
6	Withdrawn results stay in the tree, marked withdrawn.	Three early experiment artefacts carry a WITHDRAWN suffix rather than being deleted. Deleting a wrong result removes the evidence that it was found and corrected.

THE DISCIPLINE THIS PRODUCED

Seventy-three defects and failed approaches are logged, each with symptom, how it was found, root cause, fix, evidence and lesson. **Thirty of the seventy-three produced a confidently wrong answer** — the class that does not crash, does not warn, and looks entirely plausible. In a forensic system that class is the one that matters, and finding thirty of them is the reason the measurement protocol is as heavy as it is.

SECTION 02

Programme map

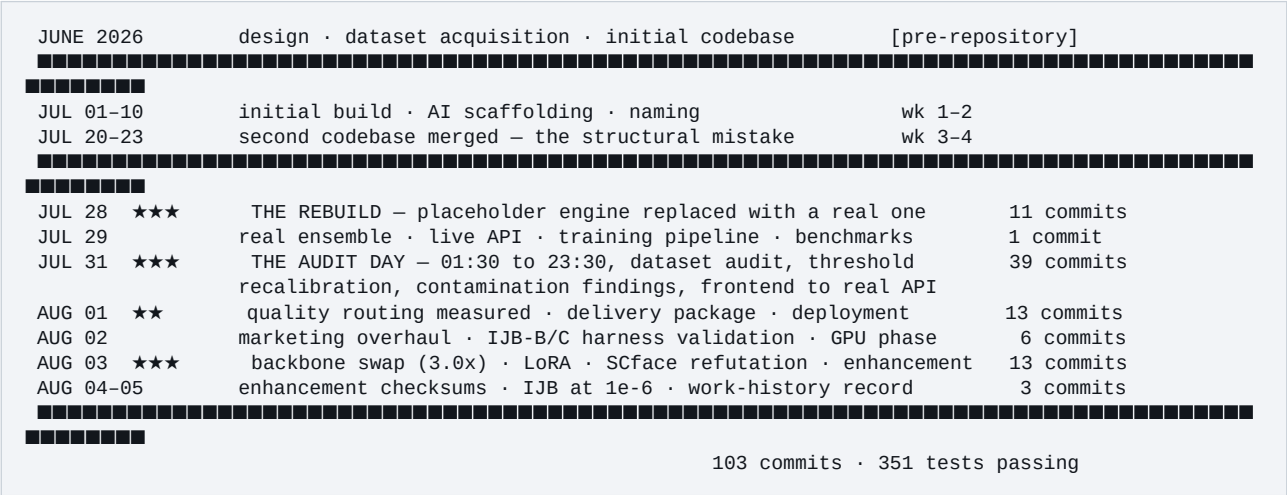


Figure 2.1 — the two months. Work is not evenly distributed: three days carry 63 of the 103 commits, and each of those three is a day on which a foundational assumption was replaced.

Aggregate	Figure	Note
Commits	103	Read in full to build this record.
Lines added / removed	348,875 / 42,707	Across 1,363 file-changes. 232,586 of the added lines (67%) are the delivery documentation, generated from the repository rather than typed.
Working sessions	18	Counted from commit timestamps; a new session begins after any long gap.
Hours worked	00:09 to 23:30	Commits land in 18 of the 24 hours. 23 of the first 98 fall between 22:00 and 06:00.
Experiments	28, with 26 artefacts	Each with its question, protocol, sample counts, confidence intervals and verdict.
Failures documented	73	Thirty of them produced a confidently wrong answer rather than an error.
Delivery documentation	12 documents, ~2,090 pages	Generated from the repository so it cannot drift away from the code.

PART A

The plan as executed

Each section states the objective set at the time, the work carried out, the gate where one applied, and the outcome — including where the outcome was a rejection.

SECTION 03

June 2026 — design and acquisition

Objective. Establish what the product is, what it must never do, and what data exists to measure it against — before writing the system that would be measured.

- Product definition: forensic 1:1 verification and 1:N identification for investigative use, distinguished from access control. The two use cases were considered and deliberately separated; serving both compromises the architecture into neither.
- Dataset acquisition: the clean protocol packs, the real degraded corpora, and the licence review each required. Registration and research-licence work begins here because it is measured in weeks, not hours.
- The initial codebase: a marketing frontend and a backend skeleton, carried into version control on 1 July as 75 files and 11,051 lines.

WHY THIS PERIOD IS RECORDED RATHER THAN SMOOTHED OVER

There are no commits in June, so it cannot be itemised day by day the way the rest of this record is. Rather than back-date or estimate it, this record states plainly what it contains and where its output enters the version-controlled history. The totals elsewhere in this document therefore reconcile exactly against git.

SECTION 04

Weeks 1–4 — initial build, and a structural mistake

Date	Work	Outcome
1 Jul	First upload: 75 files, the React/Vite frontend and a backend skeleton. Three further interface passes the same day.	Repository created at 11,051 lines. The project at this point is a website with a placeholder service behind it.
3 Jul	Large scaffolding commit: engine package layout, configuration, deployment folder, first API routes. 127 files in one commit.	Backend structure laid down. Recognition still a placeholder.
7 Jul	Boilerplate removed from the front-door documentation; project renamed.	Renamed to NexGen iMATCH.
10 Jul	Repository re-initialised.	Structural commit only.
20 Jul	A second codebase, 83 files and 12,157 lines, pushed onto the main branch and merged with a remote history.	Two unrelated products now in one repository.
21 Jul	A second 'initial commit' and a further merge of remote history.	The merge left unresolved conflict markers in three dependency manifests. Neither dependency set could be installed.
23 Jul	Documentation rewrite; a stale cross-reference removed.	

THE LESSON, RECORDED AT THE TIME

Merging a second unrelated product into a live repository cost more than it saved: broken dependency manifests, ambiguous module ownership, and a class of merge conflict that had to be resolved before any work could proceed. Both were cleared on 28 July as the first act of the rebuild. The rule that came out of it: **a repository holds one product.**

SECTION 05

The rebuild — 28 July

The single most important day of the project. Eleven commits.

The founding defect

The system could not recognise anyone. Eight separately named model 'backbones' were in fact **one class returning a hash of the image bytes as a vector**. Every score the system had ever produced was meaningless, and nothing in the interface, the logs or the tests said so: the API answered, the numbers looked like similarities, and identical images matched perfectly because identical bytes hash identically.

The work

1. Removed the obsolete second product and resolved the outstanding merge conflicts.
2. **Replaced the placeholder recognition engine with a real one** — genuine weights, a real inference session, real alignment.
3. Built the benchmark harness that could tell the difference, so the same defect could never again pass unnoticed.
4. Removed the redundant second detection pass: detection cost fell from 939 ms to 54 ms.
5. Introduced the rule that survives to today: the engine either loads real weights or raises. There is no fallback mode that keeps the API answering with meaningless numbers.

Verification

Measurement	Result	Reading
Rank-1 over 25 AgeDB identities	92.0%	The system now recognises people.
Genuine mean similarity	0.4907	A real distribution, not a hash collision artefact.
Impostor mean similarity	0.0422	Separation of 0.4485 — the property a hash-based 'engine' cannot produce.
Detection latency	939 ms → 54 ms	A redundant pass, not an optimisation.

WHY THIS IS THE FOUNDING RULE OF THE WHOLE PROGRAMME

A system that returns plausible numbers while doing nothing is worse than one that crashes. Everything in this plan that looks like excessive measurement discipline — the pre-registered gates, the reference-population purity requirement, the refusal to emit a figure without its protocol — traces to 28 July. **The defect was not that the code was wrong. It was that nothing in the system was capable of noticing.**

SECTION 06

Real ensemble, live API, training pipeline — 29 July

Phase 2 and Phase 3 in a single commit: a real three-model ensemble, live biometric routes, a fine-tuning pipeline, benchmark scripts, the frontend wired to live endpoints, and a rewritten front-door document.

Measurement	Result
Rank-1, AgeDB probes	87.32% (599 of 686)
TAR at threshold 0.28	87.60% at a false accept rate of 0.00%

Phase 2 and 3 were explicitly labelled UNVERIFIED in the same commit that delivered them. This is the rule from the previous section applied immediately: work that has been written but not measured is marked as such, in the record, at the

time — not left for a later audit to discover. The three-model ensemble delivered here was measured over the following days and subsequently rejected, on the project's own data.

SECTION 07

The audit day — 31 July

Thirty-nine commits between 01:30 and 23:30 — the longest working day in the project. Three separate assumptions were overturned.

Phase 0 — dataset audit

- Every training archive audited for identity overlap with the evaluation sets, using an embedding-based method rather than filename matching.
- **All three training archives were found contaminated.** Every fine-tuning result measured against them was therefore invalid, and was withdrawn rather than caveated.
- Contamination was reported as a floor, not a total — the audit finds what it can detect.
- Two inaccurate entries in the existing issue register were corrected as part of the same pass.

Phase 2 — the measurement items

Item	What was measured	Outcome
Latency and throughput	Per-stage timing and concurrency	Closed the 'no performance data' gap. Concurrency saturates at about four workers; batching identified as the scaling lever.
Demographic differentials	Fairness data for the <i>deployed</i> configuration, not a nearby one	Valid fairness measurement for the configuration actually shipped.
Model choice	Three-model ensemble against best single model	The ensemble does not earn its cost — it lost on four of five datasets at three times the compute. Rejected.
Exact search backend	Brute-force cost at 1k, 10k and 100k templates	Below 10■ the search is under 8% of the encode. No dependency added.
A benchmark gap	A cross-pose result below expectation	Diagnosed as a provenance problem — three different variants of the same protocol file — not an accuracy problem. Documented rather than 'fixed'.

The threshold recalibration

Two different people scored 0.2405 and the system reported the comparison as supporting the same person. The decision threshold was raised from 0.20 — the accuracy optimum — to 0.2871, the false-match-rate 0.1% operating point, measured over 40,098 pairs. False matches on hard impostors fell from 1.19% to 0.10%, **a twelve-fold reduction**, for roughly double the miss rate. The reasoning is recorded at the point of definition in the code so that anyone tempted to restore the accuracy optimum reads it first.

Phase 3 and the test suite

The frontend was wired to the real API and proven end to end. The test suite moved through 10 → 33 → 161 → 190 → 209 across the day. A fine-tune was run and measured; on contaminated data it was invalid, and it was recorded as investigated and abandoned rather than quietly dropped.

SECTION 08

Frontend to real API, batch, and the case report

Item	Work	Outcome
Items 14–18	The frontend speaks the real API	Proven end to end rather than against a mock.
Item 20	Batch endpoint, both modes	One reference against many probes became the default mode, because it is the shape investigative work actually takes.
Item 21	Forensic case report as PDF, from real search results	Not a template with fields filled in — generated from the record of an actual search.
Item 23	Claims audit against the code	An unbacked accuracy claim in the marketing copy was replaced. The wording was corrected to match the evidence, not the other way round.
—	Decision threshold raised for false-match control	0.20 → 0.2871, as above.

SECTION 09

Hygiene, adversarial input, regression gate — Phases 4 to 7

Phase / item	Work	What it found
Items 24–27	Repository hygiene; committing source files that had been written but never tracked	Files present on disk and absent from version control — a silent reproducibility hole.
Item 28	Approximate nearest-neighbour search measured	Exact search adopted; approximate rejected. Recall loss is invisible to the examiner.
Item 29	Concurrency and request batching measured	Closed the concurrency question with a number rather than an assumption.
Item 31	Per-stage latency observability	Fixed an unhandled 500 discovered while instrumenting.
Item 32	Regression gate for model, threshold and fusion changes	Makes rule 4 of Section 01 enforceable rather than aspirational.
Item 33	Adversarial input tests	Fixed a 500 and a circular import. Degenerate geometry — a 1-pixel-tall strip from a bad crop — was raising from inside the vision library and killing batch requests.
Item 35	Fusion method by condition	The winning fusion method is condition-dependent . No single method wins everywhere.
Item 36	Train/eval identity overlap audit	All three training sets contaminated.
Items 36–44	Fine-tuning line of work	Closed. Documented as investigated and abandoned, with the reasoning, rather than deleted.
Items 46, 49	Documentation audit and full test suite run	Both found real problems. An audit that finds nothing has usually not been run properly.
Items 47, 50	Issue register and final scorecard	Every original issue plus everything found since; verified working separated from known limitations.

SECTION 10

The delivery package — 1 August

Twelve documents, approximately 2,090 pages, **generated from the repository rather than written** — so they cannot drift away from the code they describe.

Document	Contents
Model card	What the model is, what it was trained on, where it fails.
A1 · Development history	Every commit in order, with the message verbatim, files and lines changed, and the measurements recorded.

Document	Contents
A2 · Evidence appendices and failure/recovery log	Each defect with symptom, detection, root cause, fix, evidence and lesson.
A3 · System architecture	Module reachability <i>computed</i> from the import graph, not asserted from a directory listing.
A4 · Model and training record	Weight checksums published.
A5 · Benchmark record	Full per-fold tables. A wrong diagnosis in an earlier draft was corrected here.
A6 · Dataset provenance	Where every corpus came from, under what licence, and what the audits found.
A7 · Threshold calibration record	The operating point and the trade it represents.
A8 · API specification	The thirty-six endpoints.
A9 · Test suite catalogue	Written while fixing an assertion counter that had been undercounting.
A10 · Security architecture	Controls and the failure each prevents.
A11 · Deployment and operations	How to run it somewhere else.

Quality-routed model selection — the day's measurement

The only unambiguous model improvement in the project, and it came from routing between two models rather than from a better one. Routing at a quality threshold of 0.539 moved TinyFace TAR@FAR=0.1% from 33.13% to 37.37% — **+4.23 points** — for a worst-case clean-set cost of 0.13 points. The routing threshold was then re-derived on data disjoint from the benchmarks, because a threshold fitted on the evaluation set is not a threshold.

THE DELIBERATE RESTRAINT ATTACHED TO THAT WIN

The routing result licenses 1:1 immediately and **does not** license 1:N. Rank ordering is more sensitive to representation drift than a threshold comparison is, so the 1:N path stays single-model until embedding compatibility is verified at gallery scale. Shipping the win everywhere it appeared to apply would have been the easy call and the wrong one.

SECTION 11

Deployment, security and account lifecycle

Work	Detail
Account lifecycle	Registration, e-mail verification by one-time code, password reset, and session revocation by epoch — backend and the matching frontend screens.
Login defects	Four stacked blockers behind a single unhelpful 'Failed to fetch' were diagnosed and fixed. Accounts that could never log in were repaired; demo footer URLs were derived from configuration rather than hard-coded.
Security headers and CSRF	Response headers applied by middleware to every response, including refusals. Double-submit CSRF on cookie-borne state changes.
Continuous integration	CI added, alongside a deployment blueprint.
Dependency trimming	Deploy dependencies trimmed to fit a constrained build limit — an exercise in knowing which dependencies are actually load-bearing.
Self-hosting	Docker Compose stack — web, API and database on one origin — as the intended shape for an on-premises forensic laboratory.
Marketing surface	Full rebuild, with the trust section rebuilt around measured evidence rather than adjectives. Main bundle 533 kB → 450 kB; home page roughly 20 screens → 10.

SECTION 12

Stage 0 — cheap falsification

Objective. Decide the project's central research bet before spending years on it, and repair the capacity measurement that everything else rests on.

The four sub-stages

ID	Work	Purpose	Status
S0.1	Embed the 153,428-image distractor set	Gives a genuine open-set 1:N benchmark with a real reference population; repairs the measurement everything else depends on	Done
S0.2	Run the full IJB-B and IJB-C suites	Validate this project's harness against the shipped reference results <i>before</i> trusting its own numbers	Done — 95.44% / 96.91% TAR@FAR=1e-4, at published ArcFace level
S0.3	Renderer-free observation-space test	The decisive experiment. Decides whether the forward-model thesis is real	Done — failed its gate
S0.4	Condition-leakage baseline	Establishes the target for the leakage penalty in later training	Done — measured at +0.078

S0.3 — the decisive experiment

The question, stated so it could only be answered yes or no: *does estimating the camera's degradation and applying that operator forward to the gallery beat simply matching resolution?* No renderer was needed — a real high-resolution image serves as the person model. Six arms:

Arm	Method
A	Embed the low-resolution probe, embed the high-resolution gallery, compare — <i>the current paradigm</i>
B1	Downsample the gallery to the probe's resolution, then compare
B2	Estimate <i>this probe's</i> point-spread function, JPEG quality and noise, apply that operator to the gallery, then compare
B2r	As B2, with the blur estimate corrected to a common pixel grid
B3	Band-limited comparison — project both sides onto the spatial-frequency band the imaging chain actually passed
C	Likelihood in pixel space under the estimated noise model

The gate, fixed in writing before the run

Pass: $B2 - B1 \geq 2.0$ points TAR@FAR=0.1% on two real degraded corpora, with bootstrap confidence intervals excluding zero. **Fail:** below 0.5 points. Between the two: inconclusive.

The result

Corpus	B2 – B1	95% CI	Verdict
SCface D1 — real optics at 4.2 m, 10,010 pairs	+0.22 points	[−0.34, +0.76]	FAIL
SCface D1 — corrected arm B2r	+0.11 points	[−0.43, +0.75]	FAIL
QMUL-SurvFace — 3,000 pairs	−4.73 points	[−8.95, −2.76]	FAIL

WHAT WAS DONE WITH A FAILED GATE

The thesis was refuted and **published as a refutation**, in its own commit, with the artefacts kept. Three earlier result files that had been produced under a flawed protocol were marked WITHDRAWN rather than deleted, so the record shows both that they existed and that they were caught. A defect found in the corrected arm — blur estimates being subtracted across different pixel grids — was fixed and the arm re-run rather than dropped. **The generative inference layer is now gated on this result, and eight weeks of work saved an estimated two years.**

The capacity measurement, broken twice and repaired

The information-capacity measurement failed twice before it was trustworthy, and both failures are instructive. The first attempt censored 99.7% of its results, because three thousand impostor pairs cannot resolve a tail below one in three thousand. The second, with a twenty-million-pair sampled pool, was invalid on a subtler ground: the assumption that a different pair implies a different identity is false. Contamination tracks identity count — 0.202% for a 568-identity corpus against 0.030% for a 5,749-identity one — and 0.2% contamination is *twice* the false-accept rate being measured at 0.1%. It made one corpus appear to fall from 96.03% to 8.40%.

The rule that came out of it is now enforced in code: **reference-population purity is the measurement, not a caveat on it**. The system refuses to emit a capacity figure without identity-labelled cohorts, rather than emitting a number with a footnote — because a number with a footnote gets quoted without the footnote.

THE RESULT THAT REFRAMED THE PROBLEM

Surveillance imagery from the QMUL corpus yields **2.92 bits** of identity information against the roughly **11.5 bits** its gallery size demands. The open-set problem on that corpus is therefore information-limited, not architecture-limited: no algorithm, present or future, can identify reliably from that imagery at that gallery size. That is a statement about physics rather than about a vendor, and it is the kind of claim the evidence layer exists to be able to make.

SECTION 13**Backbone swap, LoRA, SCface — 3 August**

The heaviest technical day since the rebuild. Thirteen commits.

The production recogniser replaced

Aspect	Detail
Change	ArcFace R50 → ViT-B KP-RPE AdaFace as the production recogniser.
Protocol	One variable changed. A 153,428-image reference population and a 50-million-pair impostor distribution.
Degraded result	TinyFace TAR@FAR=0.1%: 20.59% → 60.77% — a 3.0× gain. The largest single accuracy improvement in the project.
Clean-set check	Regression measured and accepted before adoption, not after.
Mandatory consequence	Thresholds recalibrated for the new pack rather than carried over. Recorded as a required consequence in the decision record, not left to be remembered.
Integration defects found	Three, all silent — including a channel-order mismatch and a normalisation difference between packs. All three produced plausible-looking numbers, which is what makes that class dangerous.

The LoRA fine-tune on real low-resolution data

A parameter-efficient fine-tune on genuinely low-resolution training data, run against a three-part gate fixed in advance.

Gate	Target	Result
Degraded gain	Material improvement on the target condition	PASS — QMUL TAR@FAR=0.1%: 2.31% → 4.98% → 22.35%
Clean-set regression	Within tolerance	PASS
Condition leakage	Must not increase	FAIL

Two of three gates passed. The third failed, and **the fine-tune was not shipped**. SCface — real optics at real distance — then refuted the synthetic result outright, confirming the decision. A model that improves the number by learning the capture condition rather than the identity is a model that will fail on the first camera it has not seen, and it will fail while looking confident.

Also delivered that day

- Local login and session fixes, so the workspace is reachable without an external mail round-trip.
- The enhancement module: two-track pre-processing layer, the S0.3-E experiment arms, and the investigator interface.
- The vision stage wired to write to a real database rather than to memory.
- 297 tests passing at the close of the day.

SECTION 14

The enhancement module — E0 to E10

Objective. Give an investigator a materially more useful image out of poor CCTV capture, without ever letting generated detail reach the evidential path. Investigator value first; recognition gain gated separately and strictly.

Phase	Deliverable	Gate
E0 · 2d	Types, registry, lineage, database tables, feature flag, weight cache with checksums, API smoke tests, dependency-pin regression test	The reconstructed-image type provably cannot reach the low-resolution matching path; the report drawing function cannot render an unpaired enhanced image
E1 · 4d	Analysis: motion vs defocus discrimination, anisotropic kernel, illumination, occlusion, clipping, quality assessment, plus the real-CCTV suite — infrared detection, interlace detection, true-vs-stored resolution, on-screen-display detection	Estimator accuracy measured against known ground truth, on the same protocol as the existing estimator validation
E2 · 4d	Planner plus the classical, deterministic track. CPU only, no new dependencies	Deterministic: the same input produces the same hash over 100 runs
E3 · 5d	Learned single-image stages on GPU, tiled inference with CPU fallback	A 30-pixel CCTV face produces a visibly and measurably better crop
E4 · 5d	Investigator interface: split view, synchronised zoom, plan trace with skip reasons, metric deltas, processing history, original always beside output	MILESTONE 1 — an investigator can upload a poor CCTV image and get back a significantly more useful one. Ships here, at roughly 20 days
E5 · 4d	Attribution: band-limit synthesis mask, cross-model ensemble, identity drift, wired into the interface and the report	Synthesis fraction ≈ 0 for the classical track and clearly > 0 for the generative track. If it does not separate, the method is wrong and is reported as such
E6 · 4d	Video: I-frame demultiplexing, field separation, tracking, frame selection; diffusion tier with seed-variance measurement	I-frame preference measured against a naive frame grab
E7 · 5d	Multi-frame registration and robust fusion, fuse-then-restore ordering, multi-frame agreement	Measured detail gain over the best single frame on controlled multi-frame capture
E8 · 3d	Report integration: methodology and limitations sections, audit lines, synthesis fraction and identity drift in the PDF	The report states model, version, parameters, synthesis fraction and drift

Phase	Deliverable	Gate
E9 · 1d + GPU	Recognition benchmarking — reusing the existing experiment harness rather than rebuilding it	See the pre-registered gate below
E10 · 3d	Asynchronous job queue, batch, GPU throughput tuning	p95 latency measured and published

Approximately 39 working days, with milestone 1 landing at E4. The attribution work at E5 follows immediately, because shipping generative output without the synthesis measurement would breach the standing rule that generated detail is measured rather than disclaimed.

THE PRE-REGISTERED GATE FOR E9, FIXED BEFORE ANY RUN

An enhancement stage is adopted into the *recognition* path only on $\geq +2.0$ points **TAR@FAR=0.1% with a bootstrap confidence interval excluding zero**, on a **real** degraded benchmark rather than a synthetically degraded clean one, with **no** benchmark regressing by more than 0.5 points. This is the same gate structure S0.3 used. S0.3 failed its gate and was published as a failure; this may too, and that result is publishable either way. Note that E9's outcome does not affect milestone 1 — visual examination value and recognition gain are different claims, and only the second is gated.

What this module will not do, stated in the design

- Put an enhanced image into the chain-of-custody hash or the likelihood computation.
- Present enhanced output without the original beside it and its label.
- Claim a recognition improvement before E9 measures one.
- Colourise infrared capture, or silently repair on-screen-display regions.
- Claim bit-exact reproducibility across different hardware — it defines a canonical mode instead.
- Retrain, re-threshold or otherwise modify the recogniser, or re-measure the baseline that already exists.

SECTION 15

Closing week — 4 to 5 August

Work	Outcome
The complete project work-history workbook	Thirteen sheets built from the repository: daily log, commit log, models, datasets, experiments, failures and fixes, methods tried, decisions, deliverables, pending work, and the chart data behind every figure so each chart traces to its numbers.
Enhancement weight checksums pinned; within-batch probe deduplication	Eleven-fold reduction in GPU work on realistic batches, which repeat probes.
The IJB operating point at 1e-6 recorded	With the support it actually rests on stated alongside it — an operating point quoted without its supporting pair count is not a measurement.
Backbone loaded only on a cache miss	Closed the outstanding re-run item; a warm process no longer re-pays the load.
The production ViT run on IJB; QMUL measured at 1e-6	Both distribution tails characterised — <i>what they are made of</i> , not merely where they sit.

PART B

The plan ahead

Part B is the forward plan of record: the phase order, the multi-year stage programme, the data strategy, the resourcing reality, and an explicit list of what will not be built.

SECTION 16

Phase order

Phase	Name	Gates on	Runs parallel to
8	External evaluation readiness	The processor-path accuracy and latency sweep	—
9	Ship the measured wins	Phase 8 findings	12
10	The contribution — degraded imagery	9	12
11	Submit	10	12
12	Product completeness	—	9, 10, 11
13	Publish	11	—

Phase 12 is deliberately unblocked. It is the customer-facing work — video, the evidence layer, explainability — and it must not wait on an evaluation cycle measured in months. A plan in which all product work queues behind a research milestone is a plan that ships nothing for a year.

SECTION 17

Phase 8 — external evaluation readiness

Exit criterion. A compiled library that passes the external evaluator's own validation package on the target platform, with a local harness that reproduces the submission path end to end.

#	Task	Note
8.1	Processor-path accuracy and latency sweep	Blocks everything below it. The processor path is currently unverified, and a submission binary that runs on a different execution path from the product is not a submission of the product.
8.2	Obtain and read the current evaluation API specification	Track selection follows from the specification, not before it.
8.3	Implement the required native shared-library API over the inference runtime	A thin wrapper. The engine stays as it is for the product; the submission binary is separate and shares only the weights and the preprocessing.
8.4	Bit-exact preprocessing parity harness	The submission path and the product pipeline must produce identical templates on a fixed corpus, asserted to floating-point tolerance, in continuous integration.
8.5	Template size and serialisation budget audit	512 dimensions at 4 bytes is 2 kB before any packing. Confirm against the specification.
8.6	Offline, no-network, static-link compliance check	Submission binaries run air-gapped.
8.7	Pass the validation package locally	The real gate.
8.8	Participation agreement, submission encryption, naming	Administrative, but on the critical path — start early.

THE LARGEST SINGLE UNKNOWN IN THIS PLAN

Task 8.3 is the biggest risk in Phase 8. If the native wrapper proves disproportionate, the fallback is to submit to a track with a less demanding harness first and treat it as a learning submission — which is a better outcome than a perfect submission that never ships.

SECTION 18**Phase 9 — ship the measured wins**

Everything in this phase is **already measured and not yet shipped**. It is the cheapest accuracy in the project.

#	Task	The evidence that justifies it
9.1	Ship quality-routed selection for 1:1	+4.23 points TinyFace TAR@FAR=0.1% for a 0.13-point worst-case clean cost, with the routing threshold derived on disjoint data.
9.2	Keep 1:N single-model until embedding compatibility is verified at gallery scale	Rank ordering is more sensitive to representation drift than a threshold comparison. Do not skip this.
9.3	Ship the specialist as an opt-in pack with its own version tag	It never silently replaces the default pack.
9.4	Enable the exact flat index in production	Measured; the guarded branch already exists. Needs a deliberate verification run, not merely an install.
9.5	Expose multi-face encoding through the API	Already implemented in the pipeline and never exposed.
9.6	Re-run the regression check and update the claims register	Mandatory after any model, threshold or fusion change — the register says so in its own header.

EXPLICITLY NOT DOING: MULTI-MODEL ENSEMBLING

The shipped three-model ensemble lost to the best single model on four of five datasets at three times the compute. Both candidate architecture proposals called for a six-model ensemble; this project's own data rejects it. **Reversing that decision requires new measurement, not a new diagram.**

SECTION 19**Phase 10 — the contribution: degraded imagery**

This is where the project is judged. The measured position is uncomfortable and is stated plainly: on degraded imagery, 98.7% of the total cost is discrimination loss rather than calibration loss. Better reporting cannot close it; only better recognition can. That figure is the quantitative licence for the representation-learning work, and it is also the reason the evidence layer is not presented as a substitute for it.

The strategic judgement

Be world-class at the measurement science and merely competitive at the model. The reasoning is asymmetric on purpose. Calibration in this project is already near-optimal and genuinely ahead of the field. The discrimination gap is where organisations with proprietary data and decades of investment hold structural advantages that cannot be matched with public data and a single GPU. But **nobody is competing on the evidence layer**: the capacity bound, the population model, correlated-evidence fusion, retraction-safe memory and a measured camera degradation corpus are all unclaimed, all achievable at this scale, and all things a court needs and a vendor will not build.

SECTION 20

Phase 12 — product completeness

Area	Work	Why it is in the plan
12a · Video and acquisition	Tracking, frame selection by expected information gain, correlated-evidence fusion	The largest capability gap. A face tracked across 300 frames is one observation of one person, not 300 matches — and treating it as 300 inflates the reported evidence by orders of magnitude.
12b · Forensic evidence layer	Population model, not-enrolled hypothesis, conformal candidate sets, counter-hypothesis engine	The differentiating layer; blocked on identity-labelled cohorts rather than on code.
12c · Explainable output	Why this candidate, on what regions, with what disagreement	Required for court, not optional. A conclusion an examiner cannot interrogate is not usable in testimony.
12d · Investigation intelligence	Case-level linkage and review workflow	The work an investigator actually does sits above the single search.
12e · Security and infrastructure	Remaining hardening items from the scorecard	Tracked openly in the issue register rather than closed quietly.
12f · Scale	Approximate indexing, adopted on a measured trigger rather than on principle	Recall loss is invisible to the examiner; the trigger is a gallery size, not an opinion.

SECTION 21

The stage programme

Stage	Objective	Window	Success / failure condition
S0 · Cheap falsification	Decide the generative bet; repair the capacity measurement	Weeks 1–8	Complete. The forward-operator thesis failed its gate on real optics. The generative layer is gated on it.
S1 · Upgrade what exists	Ship a materially better system on the current representation, so value lands regardless of S0's verdict	Months 2–5	Success: open-set performance beats the baseline and conformal coverage lands within $\pm 2\%$ of nominal. Failure: coverage unachievable → the population model is wrong, return to the data strategy.
S2 · Representation learning	Attack the 98.7% discrimination loss directly, with an operator-supervised objective	Months 4–14	Success: degraded TAR@FAR=0.1% $\geq 45\%$ with clean regression ≤ 0.5 points and leakage ≤ 0.02 , from 0.078. Failure: degraded gain costs more than 2 points clean → revert; the routing fallback already banks +4.23.
S3 · Generative inference	Forward rendering and likelihood in observation space	Months 12–36	Contingent on S0.3, which failed. Rung-gated: escalate only on a measured gain at the previous rung. Failure at rung 1 → stop; do not build rung 2.
S4 · Evidence integration	Capacity guard as a runtime assertion, replayable memory, retraction propagation, court report	Months 6–30, parallel	Ships value independently of S2 and S3.
S5 · Production	Tiered cascade — millisecond retrieval, second-scale refinement on the top candidates only; external submissions	Months 24–36	

The design of this ladder matters more than any individual stage. **Every stage ships value on its own**, and each later stage is gated on a measured result from an earlier one. There is no point in the plan at which two years of work must complete before anything is usable.

SECTION 22

Data strategy

A · Already held

Asset	Contents	Role
Full IJB-B and IJB-C suite	Complete: crops, metadata, protocol code and reference results	Harness validation against published results before trusting in-house numbers
TinyFace testing set	8,171 labelled images plus 153,428 gallery distractors	A real open-set 1:N benchmark with a genuine reference population
TinyFace training set	~10,400 natively low-resolution training images	Degraded-condition training data
QMUL-SurvFace	5,319 natively low-resolution identities	Real surveillance capture, under research licence
SCface	Multi-distance capture at real standoff	Real optics — the corpus that refuted the synthetic result
Clean protocol packs	Six standard verification protocols	Regression detection only; these are saturated
Cached embeddings	Seven datasets across models	Enables analysis without re-running the GPU

B · Must acquire

Surveillance benchmarks under data agreement — applications filed early because they are measured in months, not days; large-scale corpora for proposal-network pretraining; a paired face-and-voice video corpus; multi-view and three-dimensional data, which is the binding constraint on any renderer work; controlled pose-by-illumination capture for ground-truth nuisance; and longitudinal data for the aging component, where genuine decades-long data barely exists and is a real field-wide gap.

C · Must create — the differentiated assets

1. **A camera degradation corpus.** Measured point-spread function, modulation transfer, noise model and compression pipeline for the camera models that actually appear in casework. No such public corpus exists. It needs a bench and patience rather than compute, and it would be cited for a decade.
2. **Forensic reference cohorts.** Demographically declared populations for likelihood-ratio denominators. The capacity failures of Section 12 say plainly that this is the measurement.
3. **Paired high- and low-resolution same-scene capture.** The same subjects, the same moment, one high-resolution camera and one real CCTV unit at several standoff distances. This is the only data that validates forward-model fidelity against reality rather than against synthesis.

SECTION 23

Compute, storage and people

	Prototype (S0–S1)	Research (S2–S3)	Production
GPU	1 existing	8–16 datacentre-class	2–4 inference
Storage	~2 TB	50–100 TB	~10 TB
Wall-clock	8 weeks	18–30 months	6 months
People	1–2	8–15	4–6

THE CENTRAL RESOURCING FACT, STATED RATHER THAN AVOIDED

Stages 0 and 1 are genuinely achievable at current scale — and Stage 0 was, in eight weeks. **Stage 3 is not.** It needs eight to fifteen people for eighteen to thirty months. Pretending otherwise would be the most expensive mistake available: it would consume exactly the years that Stages 0 through 2 and 4 would otherwise convert into a defensible, published position. The plan is therefore deliberately asymmetric, and says so.

SECTION 24

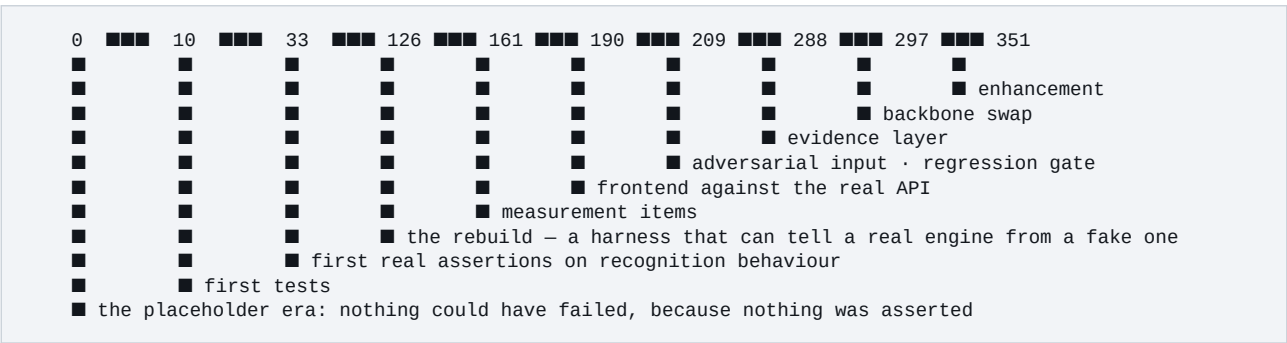
Risk register

#	Risk	Severity	Mitigation
1	Generative inference does not beat discriminative matching	Programme-ending for the generative stage	Already realised. S0.3 decided it in eight weeks, before commitment. The stage is gated and the two years are saved.
2	Forward-model mismatch produces confidently wrong evidential weights	Highest — because it is <i>silent</i>	A miscalibrated score is visibly bad; a wrong likelihood from a wrong physical model looks authoritative. Mitigated by the capacity guard, the prior-leakage null test, and calibration validated on operators the model never saw.
3	Reference-population impurity	High — already demonstrated twice	Identity-labelled cohorts only. The code refuses to emit capacity without them.
4	Data agreements denied or slow	High	Applications filed early; held corpora as the fallback.
5	Condition leaks into identity during training	High	The leakage penalty plus the measured leakage statistic as a continuous-integration gate, not a hope. This risk was realised in the LoRA run and the gate caught it.
6	Compute for the generative stage	Medium	Rung-gated escalation; no rung without a measured gain at the previous one.
7	Legal — population models edge toward ancestry inference; kinship is familial search	High	Declared, auditable strata, never inferred from the probe. Kinship is not built.
8	Team size against ambition	Highest, structurally	The asymmetric recommendation of Section 23: world-class at the measurement science, competitive at the model.

SECTION 25

Verification strategy

The test suite went from nothing to 351 passing tests over the version-controlled period, and its growth is not incidental — each step corresponds to a specific class of defect that had escaped.



What the suite is required to cover

- **Adversarial input** — degenerate geometry, empty payloads, undecodable bytes, extreme aspect ratios. Every one of these was a real 500 before it was a test.
- **Regression gating** on model, threshold and fusion changes, so a claim cannot silently outlive the measurement behind it.
- **Determinism** — the classical enhancement track must produce the same hash over 100 runs on the same input.
- **Dependency pins** — a regression test on the pinned inference stack, because a silent provider downgrade costs roughly twenty times the latency and raises nothing.
- **Chain integrity** — the audit chain verification path, including the non-deterministic ordering defect that could report false tampering.

AN AUDIT THAT FINDS NOTHING HAS USUALLY NOT BEEN RUN

Both the documentation audit and the full-suite run in Phase 7 found real problems, including an assertion counter that had been undercounting the suite's own coverage. The instrument that measures the work needs auditing as much as the work does.

SECTION 26

What will not be built

Stated in advance, so it is not drifted into later.

Excluded	Reason
A forensic-grade neural renderer in year one	It is a face world-model. Even had the falsification experiment passed, the plan goes to the cheapest rung and stops until a lab exists.
Kinship and population-substructure modelling	Scientifically interesting, legally radioactive, and it implicates people who never consented. Not built, regardless of its scientific merit.
Gait, behaviour and 'movement pattern' biometrics	The bits are not there. Gait is overclaimed in the literature and confounded by footwear, load, surface and injury; behaviour carries near-zero identity information.
Clothing and appearance matching presented as identification	That is re-identification, not identification. Presenting it as biometric would be a category error with real consequences.
Distributed vector search before 10 ¹⁰ templates	Already settled by measurement. Below that threshold the search is a single-digit fraction of the request.
Anything for airport gates or access control	Serving both forensics and access control compromises the architecture into neither.
Presentation-attack detection as a claim	The signals implemented are passive single-image heuristics and are marked uncertified. The module that would claim it is deliberately not wired in.

SECTION 27

Status ledger

Item	State	Note
Real recognition engine, benchmarked	Done	28 July. Replaced a placeholder that returned a hash of the image bytes.
Threshold calibrated for false-match control	Done	0.20 → 0.2871; a twelve-fold reduction in false matches.
Harness validated against published reference results	Done	IJB-B 95.44% / IJB-C 96.91% TAR@FAR=1e-4.
Production backbone replaced	Done	3.0× on the target degraded condition.

Item	State	Note
Quality-routed selection measured	Done, not yet shipped	Phase 9.1. +4.23 points, validated on disjoint data.
Forward-operator thesis	Refuted and published	S0.3 failed its pre-registered gate on real optics.
LoRA fine-tune	Investigated, not shipped	Two of three gates passed; condition leakage failed.
Three-model ensemble	Rejected	Lost to the best single model at three times the compute.
Delivery documentation	Done	Twelve documents, ~2,090 pages, generated from the repository.
Evidence layer — metrics and calibration	Done	Calibration is near-optimal; validated on held-out folds.
Evidence layer — population model	Blocked	On identity-labelled reference cohorts, not on code. The critical path.
Enhancement module	In progress	Through the investigator-interface milestone; the recognition gate at E9 is not yet run.
External evaluation submission	Pending	Phase 8. The processor-path sweep blocks it.
Video-native recognition	Pending	Phase 12a. The largest capability gap.
Independent validation	Not yet	Every number in this record is self-reported — accurate, protocol-stated, and self-reported. That is stated rather than glossed.

CLOSING NOTE ON THE RECORD ITSELF

Three of the largest pieces of work described in this plan ended in a rejection: the ensemble, the fine-tune, and the project's own central research thesis. All three are recorded here in the same detail as the successes, with their artefacts kept and, where results were withdrawn, marked withdrawn rather than deleted. That is the point of the record. A programme that only publishes its wins cannot be trusted on its wins.